Brain pro-resolving-to-pro-inflammatory lipids and $cPLA_2$ activation associations with late-life cognitive trajectories

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Summary

Background **Lipidome alterations are linked to neurodegenerative disorders, but the association between calcium-dependent phospholipase A2 ($cPLA_2$)-related lipid ratios, which reflect pro-resolving and pro-inflammatory lipid biology, and age-related cognitive decline remains unclear.**

Methods **In postmortem dorsolateral prefrontal cortex samples from participants in the Religious Orders Study and the Rush Memory and Aging Project (lipidomics n = 191; transcriptomics n = 411), we quantified three ratios of $cPLA_2$-released fatty acid precursors (primary exposures), three indices of arachidonic acid (AA) metabolism, and two AA transcriptomic activation scores (secondary exposures). Global cognition (primary outcome) and five cognitive domains were derived from annual neuropsychological testing.**

Findings **In mixed-effect change point models (death-age, sex, education, and apolipoprotein E ε4 (*APOEε4*)-adjusted), higher brain eicosapentaenoic acid/AA was associated with better global cognition proximate-to-death (β = 0.254, 95% CI [0.064–0.455]) and a slower rate of cognitive decline (β = 0.045, 95%CI [0.005–0.081]). Separate models including an *APOEε4* × $cPLA_2$ lipid–ratio interaction showed that the association between lipoxin A4/AA and global cognition proximate-to-death differed significantly between *APOEε4* carriers and non-carriers (β = 0.396, 95% CI [0.015–0.776]). A secondary analysis, repeating these models with three indices of AA metabolism-to-AA ratios and examining the association between the six $cPLA_2$ lipid ratios and five cognition domains, yielded similar results. Higher AA transcriptomic activation scores, indicating greater $cPLA_2$ activation, were associated with lower global cognition at death (β = −0.148, 95% CI [−0.266 to −0.028]) and with multiple cognitive domains.**

Interpretation **Modulation of brain lipid ratios warrants further investigation as a potential target related to cognitive decline.**

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Introduction

Dementia and age-related cognitive decline are rapidly growing public health challenges, with rising prevalence and substantial social and economic costs.[1] Previous studies highlighted dysregulated inflammatory processes in the central nervous system as critical drivers of neurodegeneration.[2] The brain lipidome, comprising diverse bioactive lipids, shows marked alterations in individuals with dementia and cognitive decline.[3] The apolipoprotein E ε4 (*APOEε4*) genotype, a well-established genetic risk factor for late-onset Alzheimer's disease (AD), is linked to disturbed lipid metabolism and heightened inflammation in the brain.[4]

Calcium-dependent phospholipase A2 ($cPLA_2$) is a cytosolic enzyme expressed in most cells, including neurons and glia, that regulates brain lipid metabolism

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Research in context

Evidence before this study

Lipidome alterations, including dysregulation of $cPLA_2$-dependent arachidonic acid pathways, have been linked to neurodegenerative disorders and cognitive impairment in experimental and clinical studies. However, in the context of age-related cognitive change, it remains unclear how aspects of brain pro-resolving and pro-inflammatory lipid biology contribute to late-life cognitive trajectories.

Added value of this study

In a prospective cohort study of older adults with postmortem brain lipidomic and transcriptomic assessments, a higher eicosapentaenoic acid/arachidonic acid ratio was significantly associated with better global cognition at death and slower cognitive decline, whereas a higher arachidonic acid transcriptomic activation score was significantly associated with worse cognition.

Implications of all the available evidence

Brain $cPLA_2$-related lipid ratios and activation may be relevant to cognitive decline in ageing and neurodegenerative disease, but whether they represent a viable therapeutic target remains to be determined.

by catalysing the release of the omega-6 fatty acid arachidonic acid (AA) from membrane phospholipids.[5,6] This process generates diverse bioactive lipid mediators that can either promote or resolve inflammation.[7,8] Beyond its homoeostatic roles, $cPLA_2$ is a key mediator of inflammatory and neuroinflammatory signalling and has been implicated in several brain disorders, including AD.[9,10] AA-derived mediators such as leukotriene B4 (LTB_4) and thromboxane B2 (TXB2) amplify inflammatory signalling and may contribute to the neuroinflammation seen in AD and AD-related dementias (AD/ADRD),[8] although AA itself is an essential polyunsaturated fatty acid (PUFA) required for normal cellular function.[11] In contrast, pro-resolving eicosanoids, including metabolites of omega-3 fatty acids–eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), and docosapentaenoic acid (DPA) suppress inflammation and promote tissue repair. In addition, specialised pro-resolving mediators (SPM) can be generated via AA metabolism, including lipoxin A4 (LXA_4), a potent SPM.[8,12]

A previous study of dorsolateral prefrontal cortex (DLPFC) lipidome changes in cognitive disorders found that Down syndrome and other conditions, such as autism and schizophrenia, alter the brain lipidome composition.[13] Using postmortem DLPFC samples from the Religious Orders Study (ROS) and the Rush Memory and Aging Project (MAP), we previously identified 67 lipids that were differentially associated with *APOEε4* carrier status and cognitive impairment.[14] In a smaller ROS lipidomic study, lower omega-3 fatty acids (EPA, DPA) and their neuroprotective metabolites were observed in AD and correlated with poorer cognition; *APOEε4* also interacted with elevated inflammatory and pro-resolving lipids to predict cognitive decline and plaque burden, implicating $cPLA_2$ activation in lipid dysregulation.[8] A previous clinicopathological study indicated that AD pathology, Lewy bodies, TDP-43 pathology, and vascular lesions each contribute to late-life cognitive decline and affect cognitive domains to varying degrees.[15] However, while cognitive decline is typically non-linear late in life,[15] with substantial heterogeneity in the timing and speed of accelerated decline before death,[16,17] little data exists on $cPLA_2$-related lipid ratios that reflect aspects of pro-resolving and pro-inflammatory lipid biology in cognitive decline, particularly in the context of the *APOEε4* genotype.

To address this gap, we focused on $cPLA_2$-related lipids as pathway-informative indices of AA-driven inflammatory signalling in the brain. Using ROS brain lipidomic data we constructed a $cPLA_2$-related lipid ratio panel of six measures: three ratios of pro-resolving to pro-inflammatory $cPLA_2$-released fatty acid precursors (EPA/AA, DHA/AA, DPA/AA; omega-3/AA) and three indices of AA metabolism (TXB_2/AA, LTB_4/AA, LXA_4/AA). We also used ROSMAP transcriptomic data to derive two AA activation scores. Our primary analysis associated the three pro-resolving/pro-inflammatory precursor ratios with cognitive trajectories, as measured by a global cognition score. We hypothesised that a more pro-resolving brain milieu in relation to $cPLA_2$-related lipid profile would be associated with better global cognition status and slower decline. Secondary analyses examined the three AA-metabolism indices and extended all six lipid ratios to five cognitive domains (episodic memory, semantic memory, working memory, perceptual speed, visuospatial ability). Given that *APOEε4* status and AD pathology were linked to lipid dysregulation and are contributors to cognitive decline, we further tested whether *APOEε4* status or AD pathology moderated these associations. We additionally evaluated upstream molecular regulation of AA metabolism using the transcriptomic AA activation scores.

Methods

Participants

ROS is an ongoing prospective, longitudinal, epidemiologic clinical-pathological study of ageing and dementia in older Catholic nuns, priests, and brothers from across the United States.[18] Participants without known baseline dementia agree to clinical evaluations, including detailed neuropsychological testing, each year, as well as to brain donation at the time of death. Since 1994, nearly 1600 older persons enrolled. Participants also undergo baseline blood draws (and yearly draws in a subset of participants), with serum, plasma, DNA, and white blood cells stored for research.

MAP is an ongoing community-based clinical-pathologic study of ageing that started in 1997.[18] Participants without known baseline dementia agree to clinical evaluations, including detailed neuropsychological testing, each year, as well as to donation of brain, spinal cord, nerve, and muscle at the time of death. Since 1997, more than 2600 older persons enrolled. Participants undergo baseline and annual blood draws, and have their serum, plasma, DNA, and white blood cells stored for research.

Both ROS and MAP have largely similar recruitment and data-collection strategies, enabling harmonisation of datasets. This includes the standardisation of cognitive evaluations, neuropsychologists' impressions, and diagnostic classification forms across the studies. Building on this shared framework, harmonisation has been carried out for many of the datasets across ROS and MAP, including cognitive, pathologic, omics, and other datasets, as we have published in many papers.[19–21]

Ethics
The Institutional Review Board of Rush University Medical Centre approved both studies (ROS IRB# L91020181, MAP IRB# L86121802). The studies were conducted in accordance with the Declaration of Helsinki. All participants signed informed and repository consents and an Anatomic Gift Act for brain donation at the time of death.[18]

Brain lipidome
At the time of death, a rapid brain autopsy in ROSMAP participants was performed, with a short postmortem interval, as previously published.[14] The extraction and detection of lipids from frozen postmortem brain ROS tissues from the DLPFC by liquid chromatography tandem mass spectrometry (LC-MS/MS) were performed at the University of Southern California, as detailed previously.[8] Further description is available in Supplemental Methods S1.

$cPLA_2$-related lipid ratio panel
For each lipid of interest, chromatographic peaks were manually inspected and integrated to obtain area-under-the-curve (AUC) values. These AUC values were first normalised to the weight of tissue used for lipid extraction and then further normalised to the appropriate respective deuterated internal standard to account for sample variation during processing and batch-to-batch variability. Lipid ratios were calculated from these normalised AUC values and should be interpreted as relative abundance measures.

We focused on a panel of lipid ratios indicating either pro-resolving to pro-inflammatory $cPLA_2$-released fatty acid precursors (EPA/AA, DHA/AA, DPA/AA; primary exposures) or major branches of AA metabolism (TXB_2/AA, LTB_4/AA, LXA_4/AA; secondary exposures).[8] TXB_2 and LTB_4 are products of the thromboxane and leukotriene pathways, respectively,[22,23] and were used as indices of AA flux into pro-thrombotic/pro-aggregatory and pro-inflammatory leukotriene pathways. For the lipoxin arm, we selected LXA_4 because it is a well-characterised endogenous SPM that actively promotes the resolution of inflammation.[24] Thus, we used LXA_4/AA to index the predominant pro-resolving lipoxin pathway associated with AA, while TXB_2/AA and LTB_4/AA index the major thromboxane and leukotriene branches. A higher omega-3/AA precursor ratio (EPA/AA, DHA/AA, and DPA/AA) indicates a relative shift from pro-inflammatory lipids to pro-resolving lipids in the brain. Higher TXB_2/AA and LTB_4/AA ratios indicate increased conversion of AA to pro-inflammatory mediators. In contrast, a higher LXA_4/AA ratio reflects an increased capacity for resolving inflammation and a shift towards a more pro-resolving milieu.

Transcriptomic AA activation scores
We leveraged ROSMAP DLPFC transcriptomic data to derive two complementary measures of AA pathway activity. Laboratory procedures and data processing for RNA-Seq of frozen DLPFC bulk tissues in the ROSMAP were described in previous publications[20,25] and in Supplemental Methods S2. The derived scores capture upstream regulation of AA metabolism beyond lipidomic profiles: an "AA metabolism" score (AA meta), which summarises the overall expression of genes in the AA metabolic pathway, and an "AA compass" score, which estimates the potential for AA production from phospholipids via PLA2-related reactions. A detailed description of score derivation is given in Supplemental Methods S3.

Cognitive data
Cognition was assessed annually, including using neuropsychological testing, as detailed previously.[26] Briefly, a validated battery of 19 neuropsychological tests was administered by a trained research assistant. A composite measure of global cognition, derived from all 19 cognitive tests, was calculated by converting each test's raw score to a z-score using the mean and standard deviation of the entire cohort at the baseline visit, and then averaging the z-scores. Furthermore, individual tests were grouped into five summary scores of different cognitive domains: episodic memory (7 tests), semantic memory (3 tests), working memory (3 tests),

perceptual speed (4 tests), and perceptual orientation/visuospatial ability (2 tests).

Demographics, *APOEε4* genotype, and AD and vascular pathologies

Demographic variables included age at death, self-reported sex, and years of education. *APOEε4* genotype was determined as previously described.[27] Briefly, DNA was extracted from peripheral blood mononuclear cells or brain tissue, and codons 112 (position 3937) and 158 (position 4075) of exon 4 of APOE on chromosome 19 were sequenced by Agencourt Bioscience Corporation. AD pathology was classified using the National Institute on Aging–Alzheimer's Association (NIA-AA) criteria,[28] which combine Braak neurofibrillary tangle stage, CERAD neuritic plaque score, and Thal amyloid-β plaque score. For this study, we dichotomised NIA-AA categories into AD present (high or intermediate likelihood) and AD absent (low or no likelihood). The presence of one or more chronic cerebral infarcts of any size (micro- and gross infarcts) and location (e.g., cortical or subcortical) was determined by neuropathologic evaluation, as detailed previously.[29,30]

Statistical analysis

Fig. 1 summarises the study design and analytic framework linking brain lipidomic and transcriptomic data to longitudinal cognition. Our primary analysis tested associations of EPA/AA, DHA/AA, and DPA/AA with global cognition trajectories. Secondary exploratory analyses evaluated: (1) the three AA-metabolism indices and global cognition; (2) all six $cPLA_2$-related lipid ratios and five cognitive domains; (3) effect modification by *APOEε4* status or AD pathology on lipid ratio–cognition associations; and (4) associations of two transcriptomic AA activation scores with trajectories of global cognition and the five domains. The primary and secondary exploratory analyses controlled for the same set of covariates.

Background characteristics are summarised as mean (SD) for continuous variables and n (%) for categorical variables. Pearson correlations were used for continuous variables, with a False Discovery Rate correction for multiple comparisons. We fit mixed-effects change-point models (CPMs), adjusting for sex, age at death, education, and *APOEε4*, to associate each of the six $cPLA_2$-related lipid ratios with cognitive trajectories (Supplemental Methods S4). CPMs allow the rate of change to shift at an estimated time before death, yielding the pre-terminal slope (slope 1), change-point time, terminal slope (slope 2), and intercept (cognition proximate to death).[16,17,31] Secondary analyses applied these CPM terms to each of the five cognitive domains, using: (1) a model adjusted for sex, age at

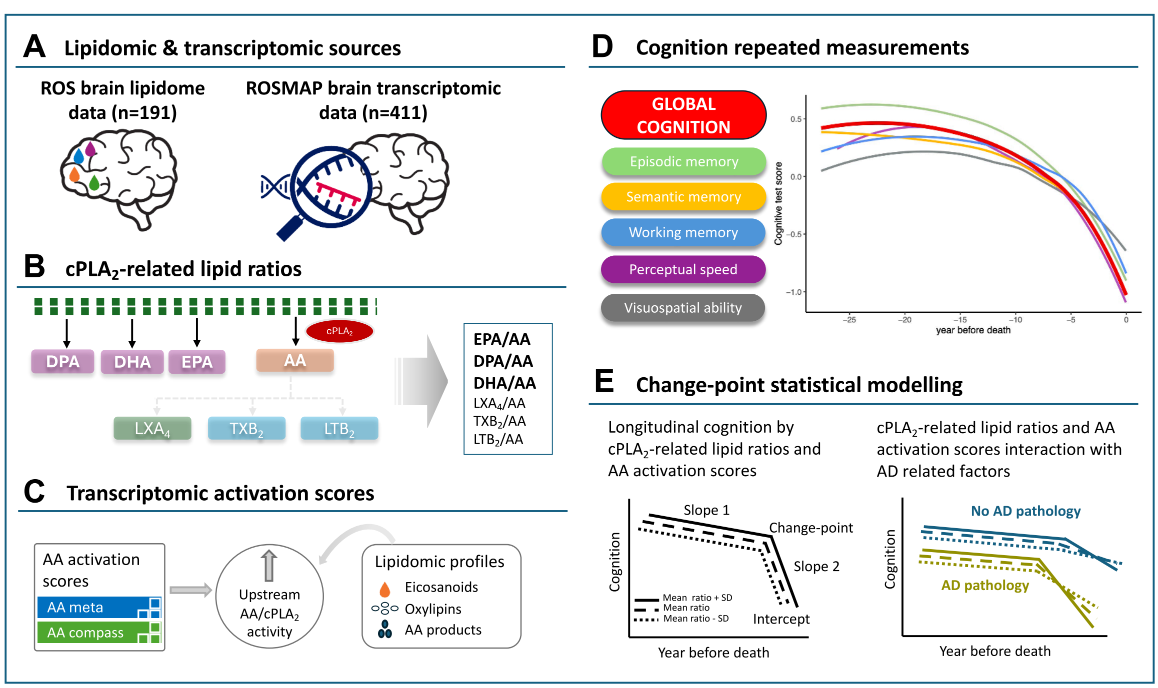

Fig. 1: **(A–E): Study Design and Analytic Framework.** A. Data source; B. $cPLA_2$-related lipid ratios; C. AA transcriptomic activation scores; D. Cognitive assessment. E. Illustration of the change-point models. AA, arachidonic acid; $cPLA_2$, calcium-dependent phospholipase A_2; DHA, docosahexaenoic acid; DPA, docosapentaenoic acid; EPA, eicosapentaenoic acid; LTB_4, leukotriene B_4; LXA_4, lipoxin A_4; MAP, Rush Memory and Aging Project; ROS, Religious Orders Study; TXB_2, thromboxane B_2.

death, education, and *APOEε4* (model 1); (2) model 1 plus APOEε4 × lipid–ratio interaction (model 2); and (3) model 1 plus the dichotomised AD pathology × lipid–ratio interaction (model 3). In an additional sensitivity analysis for the primary outcome of global cognition, we fit CPMs including interaction terms between vascular pathology (presence of ≥1 chronic cerebral infarct) and each of the three primary exposures. In parallel transcriptomic analyses, we replaced lipid ratios with each AA activation score and repeated the CPMs. Age at death and education were centred at 90 and 16 years, respectively, and scaled by 10. Analyses used R v4·5·1, SAS v9·4, and OpenBUGS v3·2·3, with two-sided α = 0.05.

Role of funders

The funding body (National Institutes of Health) was not involved in study design, data collection, data analysis and interpretation, manuscript preparation, or the decision to submit for publication.

Results

Background characteristics

One hundred and ninety-one individuals (62.3% women, mean age at death 89.4 ± 6.6 years, 18.2 ± 3.5 years of education) had brain lipidomic data and at least three annual clinical evaluations (last visit ≤ two years before death) from which cognitive data were available. There were no missing data in the main analyses; detailed sample sizes for each variable are provided in Supplemental Table S1. Among those who met the follow-up and visit inclusion criteria, the mean follow-up for those with a global cognition trajectory was 13.0 ± 5.3 years (median = 12 visits, range = 3–27, interquartile range = 8). Table 1 presents the characteristics of the 191 individuals by *APOEε4* status (carrier vs non-carrier).

$cPLA_2$-related lipid ratio panel correlation with demographics and cognition

We observed significant correlations among the $cPLA_2$-related lipid ratios. EPA/AA was positively correlated with DPA/AA ($r = 0.513$, $p = 3.38 \times 10^{-14}$), TXB_2/AA ($r = 0.200$, $p = 5.63 \times 10^{-3}$), LTB_4/AA ($r = 0.153$, $p = 0.03$), and LXA_4/AA ($r = 0.145$, $p = 0.04$), while DHA/AA was inversely correlated with LXA_4/AA ($r = -0.244$, $p = 6.83 \times 10^{-4}$). The full correlation matrix is shown in Fig. 2A, and most correlations remained significant after multiple-comparison correction (Supplemental Table S2). EPA/AA was positively correlated, and TXB_2/AA inversely correlated, with the last global cognition score proximate to death ($r = 0.171$, $p = 0.02$; $r = -0.164$, $p = 0.02$). Correlations of the lipid panel with the five cognitive domains are shown in Fig. 2B and Supplemental Table S3; none remained significant after correction. DPA/AA and LXA_4/AA were correlated with age at death ($r = -0.153$, $p = 0.03$; $r = 0.158$, $p = 0.03$), and no lipid ratio was correlated with education.

$cPLA_2$-related lipid ratio panel and cognitive trajectory

Using a CPM adjusted for age at death, sex, education, and *APOEε4* status, we examined the associations between the primary and secondary exposures and trajectories of cognitive assessment (with global cognition as the primary outcome).

A higher EPA/AA ratio was significantly associated with both a slower rate of terminal decline and higher global cognition proximate to death. Specifically, each one SD increase in EPA/AA was associated with a slower terminal decline rate by 0.045 per year ($\beta_{\text{EPA/AA on slope}} = 0.045$, 95% CI [0.005–0.081], equivalent to 12% reduction on the terminal decline rate) and with a 0.25 higher global cognition score proximate to death ($\beta_{\text{EPA/AA on intercept}} = 0.254$, 95% CI [0.064–0.455]), equivalent to 6.1 years younger age effect. Visualisation of the results for the association between the $cPLA_2$-related lipid ratios and global cognition trajectory is presented in Fig. 3A–F.

Complete results across five cognitive domains are presented in Supplemental Table S4. In summary, higher levels of certain $cPLA_2$-related lipid ratios (especially TXB_2/AA and LXA_4/AA) were associated with earlier and/or worse decline in several cognitive domains, while higher EPA/AA showed some protective effects on perceptual speed and late visuospatial decline.

Of note, lipid ratios did not, overall, appear to mediate the associations of *APOEε4* with cognitive trajectories.

$cPLA_2$-related lipid ratio panel and cognitive trajectory by *APOEε4* status

We conducted additional CPM analyses, including *APOEε4* × lipid–ratio interaction terms, to test whether lipid–cognition associations differed by *APOEε4* status. The association between LXA_4/AA and global cognition proximate to death differed significantly by *APOEε4* ($\beta_{APOE\varepsilon4*\text{LXA4/AA on intercept}} = 0.396$, 95% CI [0.015–0.776]; Supplemental Figure S1a): higher LXA_4/AA was associated with lower global cognition in non-carriers, but the opposite pattern in carriers. For most cognitive outcomes, *APOEε4* did not modify lipid–trajectory associations, although significant interactions were observed for several domains (Supplemental Table S5, Supplemental Figure S1b–g).

Brain lipid ratio mediators for $cPLA_2$ and cognitive trajectory by AD pathology

We next added AD pathology and the AD pathology × lipid ratio interaction terms to the CPMs to test whether lipid–cognition associations differed by AD status. As with *APOEε4*, most associations did not

Characteristics[a]	All (N = 191)	*APOEε4* carrier (N = 78)	*APOEε4* non-carrier (N = 113)
Demographics			
Age at death, years	89.40 (6.58)	89.10 (6.27)	89.61 (6.81)
Female sex n (%)	119 (62.3)	47 (60.3)	72 (63.7)
Education, years	18.15 (3.51)	18.26 (3.56)	18.07 (3.49)
Postmortem interval, hours	10.31 (6.93)	10.45 (6.67)	10.21 (7.12)
AD pathology n (%)			
AD not present	53 (27.7)	16 (20.5)	37 (32.7)
AD present	138 (72.3)	62 (79.5)	76 (67.3)
$cPLA_2$-related lipid ratio panel			
$cPLA_2$-released fatty acid precursors			
EPA/AA	0.030 (0.018)	0.033 (0.022)	0.029 (0.014)
DHA/AA	1.6 (0.47)	1.63 (0.52)	1.58 (0.43)
DPA/AA	0.0079 (0.0036)	0.0078 (0.0036)	0.008 (0.0037)
Indices of AA metabolism			
TXB_2/AA	0.0020 (0.0035)	0.0020 (0.0031)	0.0021 (0.0037)
LTB_4/AA	0.00062 (0.00045)	0.00061 (0.00049)	0.00063 (0.00042)
LXA_4/AA	0.00012 (0.00011)	0.00012 (0.00012)	0.00012 (0.00010)

AA, arachidonic acid; AD, Alzheimer's disease; *APOEε4*, apolipoprotein E4; $cPLA_2$, Calcium-dependent phospholipase A2; DHA, docosahexaenoic acid; DPA, docosapentaenoic acid; EPA, eicosapentaenoic acid; LTB_4, leukotriene B4; LXA_4, lipoxin A4; TXB_2, thromboxane B2. [a]Data are represented as mean (SD) unless otherwise specified.

Table 1: **Participant Characteristics by *APOEε4* status.**

vary by AD pathology. No interactions were observed for global cognition, but significant interactions emerged for visuospatial ability (Supplemental Figure S2a and b). Among participants without AD pathology, higher DHA/AA and lower LXA_4/AA were associated with lower visuospatial ability proximate to death, whereas the opposite pattern was seen in those with AD pathology.

Sensitivity analysis

To assess the influence of participants with only three cognitive assessments, we refit the three CPMs after excluding these individuals; for global cognition, two participants were excluded (Supplemental Table S6). In model 1 (Supplemental Table S7), this restriction did not materially change most associations. In model 2 (Supplemental Table S8), we observed an additional

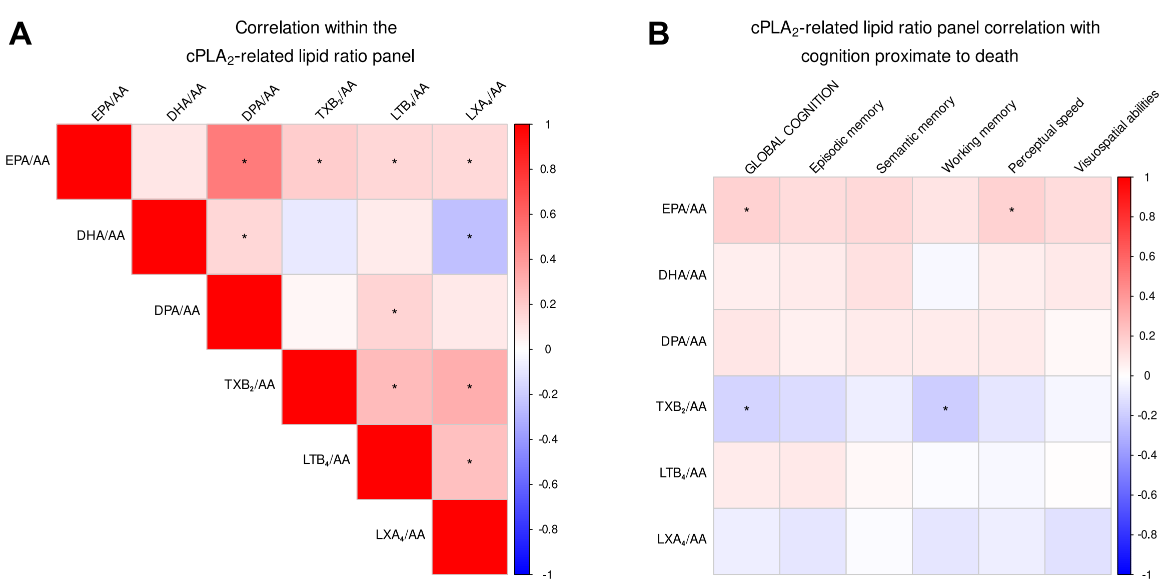

Fig. 2: **(A–B): $cPLA_2$-related lipid ratio panel correlation with cognition.** A. Pearson correlations within the $cPLA_2$-related lipid ratio panel. B. $cPLA_2$-related lipid ratio panel Pearson correlation with the last valid cognition assessment proximate to death. * Indicates significance at the 0.05 level for raw p-values. AA, arachidonic acid; DHA, docosahexaenoic acid; DPA, docosapentaenoic acid; EPA, eicosapentaenoic acid; LTB_4, leukotriene B_4; LXA_4, lipoxin A_4; TXB_2, thromboxane B_2.

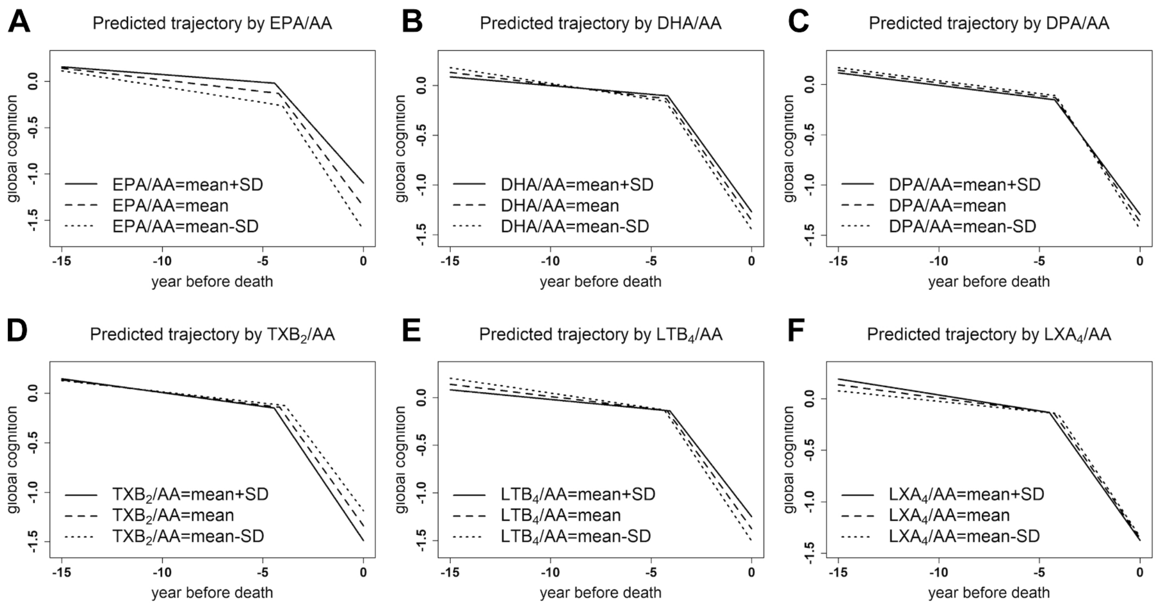

Fig. 3: **(A–F): $cPLA_2$-related lipid ratio panel and global cognition trajectory.** All models adjusted for age at death, sex, education, and APOEε4 status (n = 191). A. EPA/AA and global cognition trajectory; B. DHA/AA and global cognition trajectory; C. DPA/AA and global cognition trajectory; D. TXB_2/AA and global cognition trajectory; E. LTB_4/AA and global cognition trajectory; F. LXA_4/AA and global cognition trajectory. AA, arachidonic acid; DHA, docosahexaenoic acid; DPA, docosapentaenoic acid; EPA, eicosapentaenoic acid; LTB_4, leukotriene B_4; LXA_4, lipoxin A_4; TXB_2, thromboxane B_2.

APOEε4 × TXB2/12-HHT interaction on the perceptual speed change point ($\beta_{APOE\varepsilon4 * \text{TXB2/12-HHT on CP}} = -1.021$, 95% CI [−1.892 to −0.148]). In model 3 (Supplemental Table S9), we identified an additional AD pathology × DPA/AA interaction on the terminal slope of perceptual speed ($\beta_{\text{AD pathology*DPA/AA on slope 2}} = -0.081$, 95% CI [−0.164 to −0.001]).

We then conducted a second sensitivity analysis by adding postmortem interval (PMI) as a covariate to all CPMs. For models 1–3, PMI adjustment yielded results largely consistent with the main analyses (data not shown). In model 2, PMI adjustment revealed a significant *APOEε4* × LXA_4/AA interaction on working memory proximate to death ($\beta_{\text{APOEε4*LXA4/AA on intercept}} = 0.361$, 95% CI [0.003–0.721]). In model 3, inclusion of PMI did not alter the associations observed in the restricted sample.

In additional sensitivity analysis focussing on vascular neuropathology, we fit separate CPMs for global cognition for each of the three primary exposures, adjusting for age at death, sex, education, and *APOEε4*, and including a lipid ratio × vascular pathology (presence of ≥1 chronic cerebral infarct) interaction term. None of these interaction terms was statistically significant for any CPM parameter, indicating that the association between each lipid ratio and global cognition trajectories did not differ by the presence of chronic cerebral infarct.

Transcriptomic AA activation scores and cognitive trajectories

Finally, we replaced the $cPLA_2$-related lipid ratios with the two transcriptomic AA activation scores (AA meta, AA compass) in all three CPMs to test whether upstream AA-pathway gene expression could explain or extend the lipid-based findings, using a larger ROSMAP sample with transcriptomic data (n = 411; Supplemental Table S10; Supplemental Figure S3). For the global cognition primary outcome, the mean follow-up was 8.0 ±3.6 years (median = 8 visits, range = 3–18, interquartile range = 5). Of note, 76 individuals had both lipidomic and transcriptomic data, with age at death and sex distribution comparable to those of individuals with only lipidomic or transcriptomic data. Neither score mediated the effect of *APOEε4* on cognitive trajectories. AA meta was associated with an earlier change point and a faster pre-change-point decline in working memory, and higher AA compass was associated with lower cognition at death (global cognition, episodic memory, working memory, perceptual speed; model 1, Supplemental Table S11). No *APOEε4* × AA score or AD pathology × AA score interactions were detected (models 2–3; data not shown).

Discussion

In this study, the ratios of some pro-resolving and pro-inflammatory $cPLA_2$-released fatty acid precursors and indices of AA metabolism were associated with global cognition. We further showed that these lipid ratios were also associated with the levels and trajectories of most cognitive domains. Among the precursor ratios, associations were most consistent for EPA/AA, whereas DHA/AA and DPA/AA showed less consistent associations with cognitive trajectories. We also observed interactions between $cPLA_2$-related lipid ratios and *APOEε4* and AD pathology, indicating that these factors modify lipid-related cognitive trajectories.

Lipids are critical in regulating brain function and neurologic state.[32] Our findings indicate that a higher pro-resolving-to-pro-inflammatory fatty acid precursor ratio was associated with better cognitive status and slower terminal cognitive decline. These findings are consistent with evidence that omega-3 fatty acids confer neuroprotection via anti-inflammatory and pro-resolving pathways in mild cognitive impairment.[33] Although AA itself has been linked to slower cognitive decline and lower dementia risk in some studies,[34] others report better cognition with higher serum omega-3/omega-6 ratios and lower omega-6/omega-3 intake ratios[35,36] and overall findings remain mixed.[37] TXB_2/AA, a marker of thromboxane pathway activity,[38,39] showed negative associations with cognition, suggesting that enhanced AA-driven pro-thrombotic and inflammatory signalling may accelerate cognitive deterioration, consistent with observed links between thrombosis and AD.[40] Although LXA_4 is considered a pro-resolving mediator,[8,38,39] a higher LXA_4/AA ratio was also associated with worse cognition, underscoring the complexity of resolution pathways in advanced ageing and neurodegeneration.[41] The apparently adverse associations of LXA_4/AA should be interpreted with caution, as higher LXA_4 levels may reflect compensatory activation of resolution pathways in response to greater underlying inflammation.[41] Disease stage and other unmeasured factors (e.g., more pro-resolving lipids in MCI, more pro-inflammatory lipids in dementia) may further shape these associations and could help explain the absence of associations between $cPLA_2$-related lipid ratios and episodic memory, the domain typically affected earliest in AD. Consistent with their shared enzymatic origin and common measurement in the same brain region, the $cPLA_2$-related lipid ratios were correlated with one another, reflecting a relatively coherent pathway-informative signal. By contrast, late-life cognitive performance is a distal, multifactorial phenotype that integrates mixed neuropathologies, life-course exposures, cognitive resilience, and measurement error in cognitive testing, with $cPLA_2$-related lipid signalling as only one component. As a result, strong internal correlations among $cPLA_2$-related lipid ratios do not necessarily translate into broad or large associations with cognitive test performance in this heterogeneous ageing cohort.

Given that *APOEε4* integrates genetic risk for Alzheimer's disease with disruptions in lipid homoeostasis,[4] we assessed whether *APOEε4* status modified the associations between pro-resolving lipid mediators and cognitive trajectories. Although *APOEε4* status did not independently influence cognitive trajectories in our primary models, significant interactions between *APOEε4* and specific lipid mediators, such as LXA_4/AA, were observed. These findings highlight the complexity of lipid metabolism in the context of genetic risk for Alzheimer's disease and cognitive decline.[42] This interaction suggests that *APOEε4* carriers could benefit from interventions that enhance pro-resolving lipid pathways. However, further mechanistic studies are needed to clarify whether these effects are causal or reflect adaptive responses to chronic inflammation.

Dementia reflects advanced neurodegenerative and inflammatory processes.[43] While AD pathology status did not substantially modify the association between $cPLA_2$-related lipid ratios and cognitive trajectories, we observed that in individuals without AD pathology, higher DHA/AA and lower LXA_4/AA were associated with poorer visuospatial ability at death. In contrast, among those with AD pathology, the directions of these associations with visuospatial ability were reversed. This pattern suggests that the cognitive correlates of the pro-resolving-to-pro-inflammatory ratio differ between AD and non-AD brains and may reflect distinct underlying pathophysiological processes in these groups. Although our observational design precludes causal inference, our results underscore the importance of accounting for disease when evaluating lipid-based interventions. Also, our results suggest that strategies aimed at modulating lipid ratios, which reflect pro-resolving and pro-inflammatory lipid biology, may be most effective before the onset of neurodegeneration.[44] Nevertheless, our interaction analyses were exploratory and should be interpreted with caution, given that they were additional analyses (not the aim of the study), involved multiple testing, and had limited power within subgroups.

Whereas lipid ratios capture downstream products of $cPLA_2$ activity, transcriptomic scores reflect the coordinated expression of genes involved in AA metabolism, providing a complementary view of AA-related inflammatory signalling. Prior work found dysregulated AA metabolism and its eicosanoid products in neuroinflammation and cognitive decline, including in dementia.[45,46] Increased $cPLA_2$-mediated liberation of AA from membrane phospholipids can fuel cyclooxygenase- and lipoxygenase-dependent production of pro-inflammatory eicosanoids, amplifying microglial and astrocytic activation, oxidative stress, and synaptic dysfunction.[47] A previous study that leveraged a large patient transcriptomic dataset to inform lipid metabolism in AD[48] identified lipid metabolism-related differentially expressed genes in the hippocampus and blood of AD patients. They further found a correlation between the dysregulation of lipid metabolism, immune activity, and cellular pathophysiology. In our study, the AA activation scores provide indirect transcriptomic evidence consistent with higher AA/$cPLA_2$-related pathway activity. The transcriptomic analyses should be viewed as complementary, pathway-informative evidence supporting the lipidomic findings, rather than as within-person multi-omics validation.

Our study has several limitations. First, we focused on a pre-selected panel of lipid ratios rather than the full lipidome. However, these ratios capture biologically meaningful aspects of pro-resolving to pro-inflammatory precursors, as well as indices of AA metabolism central to neuroinflammatory regulation and cognitive ageing. Second, the use of postmortem brain tissue and an observational design limits inference about dynamic changes over time, raises the possibility of reverse causation (whereby late-life cognitive decline and neuropathology influence brain lipid profiles and $cPLA_2$-related pathways), and precludes causal conclusions or firm predictions about the effects of modifying these ratios. Yet, the use of postmortem tissue enabled region-specific measurement of brain lipids that cannot be assessed directly in vivo, and linking these profiles to well-characterised longitudinal cognitive data strengthens the relevance of our findings to ageing and dementia. Also, because lipidomic profiling was performed in postmortem tissue, altered lipid ratios may reflect downstream consequences of disease processes or neurodegeneration, rather than causal drivers of cognitive decline. Third, although we examined the presence of chronic cerebral infarct as an indicator of cerebrovascular pathology in secondary analyses, we did not characterise the full spectrum of cerebrovascular lesions or other common late-life neuropathologies such as Lewy body disease or TDP-43 pathology; these additional pathologies may contribute to mixed disease processes and residual confounding of the observed associations. Fourth, we relied on bulk RNA-seq, which does not distinguish contributions from neuronal, astrocytic, microglial, endothelial, or infiltrating immune cells to AA pathway activation. Fifth, our CPMs assume piece-wise linearity, while age-related cognitive decline may follow nonlinear patterns over time. Yet, previous studies have shown that associations between neuropathologies and cognitive decline are consistent across linear and nonlinear models,[15] and CPMs provide a useful framework for characterising how risk factors relate to both the onset of terminal decline and the rates of decline before and after its onset.[17,31] Sixth, although some information on medication use was available, a detailed modelling of specific drug classes (e.g., COX inhibitors) was beyond the scope of the present analyses, and we cannot fully exclude medication-related residual confounding. Also, residual confounding from other unmeasured factors and lack of replication in an independent cohort remain concerns. Finally, because lipidomic and transcriptomic profiling were performed in only one brain region (DLPFC), findings may not generalise to other brain regions involved in cognitive decline. Generalisability is also limited since the analytic sample consisted of older research volunteers who consented to brain donation at the time of death and does not fully represent the broader ageing population. Nevertheless, the strengths of this study include postmortem tissue, which allowed region-specific measurement of DLPFC lipids not accessible in vivo; combined with summary scores from repeated cognitive assessments and DLPFC transcriptomics, these data uniquely link brain lipid ratios and AA metabolism with cognitive trajectories in late life.

In conclusion, this longitudinal study of older deceased individuals indicates that brain pro-resolving to pro-inflammatory $cPLA_2$-derived lipids are linked to cognitive trajectories, suggesting that lipid metabolism may be relevant to developing strategies to mitigate cognitive decline. Yet, the lipid ratios should be interpreted as pathway-informative indices of relative lipid abundance, not direct measures of enzymatic flux. The interplay among pro-resolving to pro-inflammatory $cPLA_2$-released fatty acid precursors and diverse branches of AA metabolism may be a determinant of neuroinflammatory status and cognitive function in ageing. Future studies of brain AA-related lipid metabolism in late-life cognitive decline should test whether modifying these lipid ratios through dietary, pharmacological, or behavioural interventions, and using peripheral lipid profiles as less-invasive proxies for brain lipid balance, can alter neuroinflammation and improve cognition, while accounting for individual variability in factors such as *APOEε4* status and the timing of intervention.

Contributors

Z.A. and H.N.Y. conceptualised the study design. D.A.B., H.N.Y., and Z.A. obtained funding. D.A.B and Z.A. provided data and biospecimens. X.W. analysed the data. B.L. calculated the transcriptomic activation score. A.Y.M. wrote the first draft of the paper, incorporated revisions, and finalised the paper for publication. A.Y.M., X.W., B.L., R.S.W., D.A.B, S.G.L., H.N.Y., and Z.A. contributed to the final version of the paper. A.Y.M., X.W., D.A.B, H.N.Y., and Z.A had full access to all the data in the study. All authors read and approved the final version of the manuscript. All authors accept responsibility to submit for publication.

Data sharing statement

The Rush Alzheimer's Disease Centre (RADC) in Chicago, Illinois, has a long track record of data sharing. Data from the Religious Orders Study (ROS), Rush Memory and Aging Project (MAP), and other RADC data can be requested by submitting a brief online data request form. After committee review, approval, and signing of a Data Use Agreement, data will be released. Researchers should visit the RADC Research Resource Sharing Hub at: https://www.radc.rush.edu. Omics data should be requested from the ADKnowledge portal synapse.org.

Declaration of interests

Zoe Arvanitakis receives research funding from the National Institutes of Health (NIH), the Illinois Department of Public Health (IDPH), and industry to conduct clinical trials. Zoe Arvanitakis serves as a consultant for national and international grant review agencies (including the

National Institutes of Health, California Institute of Regenerative Medicine and European governments, among others). Zoe Arvanitakis consults for not-for-profit and for-profit companies, including Eisai, Novo Nordisk, and Summus Inc. Zoe Arvanitakis is an expert consultant for law firms working on medico-legal cases. Zoe Arvanitakis is on the Editorial Board of the scientific journal Frontiers in Dementia. Hussein N. Yassine and Stan Louie are the founders of Pebrx Corp. All other authors have no conflicts.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ebiom.2026.106421.

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